Tade Marozsak

PhD Candidate at MIT EECS

He/Him
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Education

Massachusetts Institute of Technology, *EECS, PhD* 2024 –

GPA 5.0
Developing advanced optical imaging and computational methods for the non-invasive, high-contrast visualisation and analysis of biosystems in Prof. Sixian You's Biophotonics Laboratory. I research using multiharmonic-generation imaging for sub-resolution tissue characterisation and modelling high-refractive-index lipids.

University of Oxford, *Engineering Science, MEng* 2020 – 2024

Awarded First Class Honours each academic year, with an overall First Class degree
Courses in optoelectronics, machine learning, medical imaging, and communications. Beyond my university commitments, I tutored socioeconomically challenged students, and served as the college ice hockey captain.

Awards

Thomas and Sarah Kailath Fellowship
Massachusetts Institute of Technology 2024 – 2025

Motz Prize (Best Project in Electrical Engineering)
Dept. of Engineering, University of Oxford *Aug.* 2024

Clarendon Fund Scholarship (declined for MIT)
Council of the University of Oxford *Apr.* 2024

Casberd Scholarship (ranked 7th in exams)
St John's College, University of Oxford 2021 – 2024

Honourable Mention, European Physics Olympiad
EuPhO Academic Committee *Jul.* 2020

1st Place, National Physics Olympiad
Hungarian Educational Authority *Apr.* 2020

Publications

A reconfigurable arbitrary retarder array as complex structured matter 2025
Nature Communications — *Chao He and others (including Tade Marozsak)*

Multiplexed vector beam conversion via complex structured matter 2025
Nature Communications (accepted, minor revisions) (*Co-first authors) — *Runchen Zhang*, Tade Marozsak*, An Aloysius Wang* et al.*

Topological protection of optical skyrmions through complex media 2024
Light: Science & Applications — *An Aloysius Wang and others (including Tade Marozsak)*

Research

Mueller Matrix Polarimeter Uncertainty Quantification *Oct.* 2023 – *Jun.* 2024
Vectorial Optics and Photonics & Soft Matter Photonics Research Groups, Oxford — Supervisors: Dr Chao He, Prof Stephen Morris, Prof Steve Elston

- Proposed methodologies to mitigate and assess the impact of random and systematic errors in polarimetric imaging, enabling accurate diagnostics of subtle spatially varying optical properties.
- Established an approximate upper bound on the uncertainty induced by random errors in Mueller matrix elements and their derived parameters.
- Improved parameter accuracy through numerical optimisation techniques, and estimated the residual error using bootstrapping to provide combined uncertainty estimates for random, systematic, scattering, and diffraction errors.

Paraxial Skyrmion Beam Generation *Jun.* – *Sep.* 2023

Vectorial Optics and Photonics & Soft Matter Photonics Research Groups, Oxford — Supervisors: Dr Chao He

- Autonomously developed a class of passive optical systems, acting as input-dependent polarization state converters that possess the capability of generating topologically distinct paraxial beams.
- Contributed to three publications on optical Skyrmions and universal modulators, including a first-authored paper on passive optical generation.

Beam Profile Monitor for CERN *Oct.* 2022 – *Jun.* 2023

Department of Engineering Science, Oxford — Supervisors: Dr Daniel Eakins, Dr Gianfelice Cinque

- Led a four-member team in designing a beam halo monitor for the High-Luminosity Large Hadron Collider; delivered group presentations to CERN scientists, and co-authored a comprehensive 120-page technical report.
- Proposed a combined transition radiation and scintillation screen design after evaluating various approaches to overcome signal detection challenges and physical limitations, which was validated through numerical simulations.

Transport Properties of Light Species *Jul.* – *Sep.* 2022

Numerical Analysis & Hypersonics Research Groups, Oxford — Supervisors: Dr Luca di Mare, Prof Matthew McGilvray

- Awarded the highly selective EUROP scholarship (1 of 25 students) to simulate the transport properties of atomic hydrogen via Chapman-Enskog theory.
- Incorporated quantum effects by extracting collision cross sections via direct integration of the Schrödinger equation and the JWKB method, and implemented numerical methods based on self-taught theoretical background.

Teaching, Leadership & Outreach

Teaching Fellow, Deep Learning, *MITx MicroMasters* *Sep.* 2026 –
Lead recitations and office hours; contribute to editing course materials and producing studio recordings.

Social Chair, *MIT EECS Graduate Student Association* *Jan.* 2026 –
Organize department-wide programming for an inclusive environment with broad student engagement across diverse labs.

Executive Board Member & Mentor, *MIT EECS Graduate Application Assistance Program* 2024 – 2026
Responsible for mentor-mentee matching and mentoring to support underrepresented groups with their grad applications.

Mathematics Tutor, *The Access Project* 2023 – 2024
Weekly tutoring of socioeconomically disadvantaged students, reciprocating the IPhO preparation classes I benefited from.